

Shri G. S. Institute of Technology and Science, Indore
Department of Applied Mathematics and Computational Science
B.Tech. 1 Year (Group A: BM, CE, CS, EC, EE) (July-December 2025)
Mid Term Test-I MA10021: Mathematics for Engineers

Max. Marks: 20

Time: 1 hour

Note: Attempt all questions. All questions carry equal marks.

S.No	Questions	Marks	CO	BL
Q.1	(a) Write the contribution in mathematics by Baudhayana.	02	CO1	1,2
	(b) If $u = xyz$, $v = x^2 + y^2$ and $w = x + y + z$, then find the Jacobian $\frac{\partial(x,y,z)}{\partial(u,v,w)}$.	03		
Q.2	Discuss the maximum or minimum values of the function $f(x, y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$.	05	CO1	2
Q.3	Solve the differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = 2e^{2x}x + 3e^x \cos 2x$.	05	CO4	2
Q.4	Find the solution of differential equation by method of variation of parameter $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = e^x \sin x$.	05	CO4	2
